Fabian N. Monrose

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Last update: June 18, 2026

Education

Ph. D., Computer Science, *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1999
Advisor: Prof. Zvi Kedem

M. Sc., Computer Science, *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1996

B. Sc., Computer Science, *Barry University*
Miami, Florida, USA — May, 1993

Professional Experience

Julian T. Hightower Chaired Professor, *Georgia Tech*
College of Engineering, — August 2022 — present

Kenan Distinguished Professor, *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2017 — July 2022

Co-Founder, *Zeropoint Dynamics*
Role: Chief Scientist, — March 2015 — 2022

Dir., Computer & Information Security *Renaissance Computing Institute*
Joint appointment with UNC-CH Comp. Sci. — January 2014 — 2018

Professor, *University of North Carolina, Chapel Hill*
Computer Science Department, — January 2013 — present

Associate Professor, *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2008 — December, 2012

Associate Professor, *Johns Hopkins University*
Computer Science Department, — July 2007 —June 2008

Assistant Professor, *Johns Hopkins University*
Computer Science Department, — November 2002 — June 2007

Research Scientist, *Bell Laboratories, Lucent Technologies*
Secure Systems Research, — July 1999 — October 2002

Recent Honors

- Distinguished Reviewer, *ACM CCS Symposium*, — Fall, 2025
- CIOS Honor Roll, *Center for Teaching and Learning*, — Fall, 2023
- CSSA Graduate Teaching Award, *CS Department*, — May, 2021
- Undergraduate Students Teaching Award, *CS Department*, — May, 2015

- Best Student Paper Award, *IEEE Symposium on Security & Privacy* May, 2013
- Outstanding Research in *Privacy Enhancing Technologies* (PET), July, 2012
- AT&T Best Applied Security Paper Award, *NYU-Poly CSAW*, Nov, 2011
- Best Paper Award, *IEEE Symposium on Security & Privacy*, May, 2011
- Best Paper Award, *Intl. Conf. on Internet Monitoring and Protection*, April, 2011
- Faculty Research Award, *Google* May, 2011

Funding Awarded Grants & Contracts (*last 15 years only*)

PI, DARPA (Jim Simpson (*lead, Cynnovative*)), STTR Phase II: Learning an Unknown Compiler for Instant Development (LUCID) for $540,000. April 2026 – May 2028.

Co-PI, ARPA-H (Brendon Saltaformaggios (*lead*)), Hospital-Integrated Vulnerability Identification and Proactive Remediation (H-VIPER) for $208,000. October 2025 – May 2027.

Co-PI, NAVY/National Security Technology Accelerator (K. Bittick at GTRI (*lead*)), Firmware Bill of Materials Extractor for $684,000. August 2024 – July 2027.

PI, DARPA (M. Antonakakis at GaTech (*lead*)), A Novel Framework to Assess, Statistically Model and Evade Cyber Attack Infrastructure for $1,266,714, October. 22 — October 2025.

Co-PI (with Jan-Michel Frahm), CIA, Cyber Identity and Behavioral Analytics Research Consortium (CIBAR) for $559,000, July 2017 — Dec, 2021.

PI, DARPA (M. Antonakakis at GaTech (*lead*)), Attributing Cyber Actors through Tensor Decomposition and Novel Data Acquisition for $2,019,181, Nov. 2016 — August 2021.

PI, NIST, Knowledge is Power: Assessing the Security and Privacy of Consumer IoT Architectures and Systems for $183,063.00, July 2017 — July, 2018.

PI, IEEE, Advancing Cybersecurity Education Through Active, Challenge-Based, Learning Exercises for $119,999.44, Jan. 2017 — August, 2018.

PI, Department of Defense University Research Instrumentation Program (DURIP), Next Generation Defenses against Web-based Exploits for $116,500.00, April, 2016 — May 2017.

Co-PI (Jan-Michel Frahm (*lead*)), Department of Defense University Research Instrumentation Program (DURIP), Understanding Privacy Risks of Ubiquitous Personal Augmented Reality Head-mounted Displays for $95,385.00, June, 2014 — May 2015.

PI, NSF *Secure and Trustworthy Computing*, TWC: TTP Option: Small: Scalable Techniques for Better Situational Awareness: Algorithmic Frameworks and Large Scale Empirical Analyses for $667,900.00, Sept. 2014—August 2017.

PI Department of Homeland Security *Transition to Practice Program*, Supplement to NSF SDCI Sec: New Software Platforms for Supporting Network-wide Detection of Code Injection Attacks for $350,000.00, Sept. 2014—August 2015.

PI, Department of Defense University Research Instrumentation Program (DURIP), Next-generation defenses for securing VoIP communications for $114,345.00, June, 2013 — May 2014.

PI, NSF *Secure and Trustworthy Computing*, Toward Pronounceable Authentication Strings for $499,997.00, Aug. 2013—July 2016.

PI, Department of Homeland Security, Efficient Tracking, Logging, and Blocking of Accesses to Digital Objects (C. Schmitt and M. Bailey) for $1,035,590. September 2012 — August 2014.

Co-PI (Jan-Michel Frahm (*lead*)), NSF *Division of Information & Intelligent Systems*, EAGER: Automatic Reconstruction of Typed Input from Compromising Reflections for $151,749.00, August 2011—July 2013.

PI, *Verisign Labs Research Awards Program*, On Exploring Applications of Phonetic Edit Distance for $65,000.00, June 2011.

PI, *Google Faculty Research Awards Program*, ShellOS: An Efficient Runtime Platform For Detecting Code Injection Attacks for $61,635.00, May 2011.

PI, NSF *Office of Cyberinfrastructure*, SDCI Sec: New Software Platforms for Supporting Network-wide Detection of Code Injection Attacks for $800,000.00, Aug. 2011—July 2014.

PI, NSF *Trustworthy Computing*, TC: Small: Exploring Privacy Breaches in Encrypted VoIP Communications for $496,482.00, Aug. 2010—July 2013.

PI (A. Stavrou at GMU (*lead*)), NSF *Trustworthy Computing*, Scalable Malware Analysis using Lightweight Virtualization for $259,264.00, Sept. 2009—August 2011.

Manuscripts

What Typos Don't Explain: An Empirical Study of Queries of Non-Existent Domain Names. Boladji Vinny Adjibi, Michael Bailey, and Fabian Monrose. Under review at the Internet Measurement Conference (IMC), 2026.
Measurement Systems Security AI/ML

Beyond Detection: Source Attribution of Synthetic Speech using Multilabel Classifier Chains. Boo Fullwood and Fabian Monrose, under revision for re-submission to Network and Distributed Systems Security (NDSS), 2027.
AI/ML Systems Security Measurement

Acoustic Microcues: Breath and Phoneme-Level Signals for Unmasking Audio Deepfakes. Lydia Han, Amogh Palasamudram, Boo Fullwood, Fabian Monrose. In progress, 2026.
Biometrics AI/ML Measurement

Exploiting Software-level Abstractions to Support Practical Hardware Trojan Attacks. Athanasios Moschos, Georgios Kokolakis, Kevin Valakuzhy, Fabian Monrose and Angelos Keromytis, in revision for submission to Symposium on Hardware Oriented Security and Trust (HOSTS), 2027.
Program Analysis Malware Measurement

Publications
(*last 15 yrs only*)

BRAID: Uncovering Malware Family Relationships in Data-Scarce Settings. Kevin Valakuzhy, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
Malware Systems Security Forensics

HeartBreaker: Practical Attacks for Recovering Cryptographic Keys Derived from Biosignals. Evangelos Froudakis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
Privacy Biometrics AI/ML

The Impact of Emerging AI Practices on the Cybersecurity Workforce. Miuyin Wong, Alan F. Luo, Yunzhe

Zhao, Fabian Monrose, and Michelle Mazurek. USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2026.
Human Factors Privacy Policy

Beyond the Stars: Multimodal Detection of Scams on GitHub. Tillson Galloway, Kevin Valakuzhy, Manos Antonakakis and Fabian Monrose. USENIX Security Symposium, August, 2026.
Supply Chain Systems Security AI/ML

SoK: Multi-Layer Indirect Call Analysis in the Real World. Yufei Du, Vasileios P. Kemerlis, Michalis Polychronakis and Fabian Monrose. USENIX WOOT Conference on Offensive Technologies, August, 2026.
Program Analysis Systems Security Software Systems

The Cost of Silence: Quantifying the Prejudice of Non-Participation in Domain Name Disputes. Boladji Vinny Adjibi, Kathryn Kleiman, Michael Bailey, and Fabian Monrose. Int. Conference on Artificial Intelligence and Law (ICAIL), June, 2026.
Policy Measurement AI/ML

LUMEN: A Systems Approach to LLM-Guided Activation of Hidden Behaviors in Malware. Kevin Valakuzhy, Miuyin Yong Wong, Doug Blough, Mustaque Ahamad and Fabian Monrose. International Conference on Dependable Systems and Networks (DSN), June, 2026.
Systems Security Malware Forensics

Mind Your [M]s, Cross Your [T]s: A large-scale Phonemic Analysis of Speech Reproduction in Modern Speech Generators. Boo Fullwood and Fabian Monrose. IEEE Conference on Acoustics, Speech and Signal Processing (ICASSP), May 2026.
Acoustics Privacy AI/ML

Actively Understanding the Dynamics and Risks of the Threat Intelligence Ecosystem. Tillson Galloway, Allen Chang, Omar Alrawi, Thanos Avgetidis, Manos Antonakakis and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
Measurement Supply Chain Systems Security

Repairing Trust in Domain Name Disputes Practices: Insights from a Quarter-Century's Worth of Squabbles. Boladji Vinny Adjibi, Athanasios Avgeditis, Manos Antonakakis, Alberto Dainotti, Michael Bailey and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
Policy Measurement AI/ML

From Concealment to Exposure: Understanding the Lifecycle and Infrastructure of APT Domains. Athanasios Avgetidis, Boladji Vinny Adjibi, Panagiotis Kintis, Tillson Galloway, Zane Ma, Omar Alrawi, Manos Antonakakis, Roberto Perdisci, Fabian Monrose and Angelos Keromytis. International Symposium on Research in Attacks, Intrusions and Defenses, October, 2025.
Measurement Systems Security AI/ML

The Guardian of Name Street: Studying Defensive Registrations for the Fortune 500. Boladji Vinny Adjibi, Athanasios Avgetidis, Michael Bailey, Manos Antonakakis and Fabian Monrose. In Network and Distributed Systems Security (NDSS), Feb. 2025.
Measurement System Security AI/ML

Seeing is Believing: Interpreting Behavioral Changes in Audio Deepfake Detectors Arising from Data Augmentation. Boo Fullwood and Fabian Monrose. In 18th ACM Workshop on Artificial Intelligence and Security, October, 2025.
AI/ML Forensics System Security

Understanding LLMs Ability to Aid Malware Analysts in Bypassing Evasion Techniques. Miuyin Wong, Kevin Valakuzhy, Mustaque Ahamad, Douglas M. Blough, and Fabian Monrose. In Int, Conference on Multimodal Interaction (ICMI), 2024.
Malware AI/ML Forensics

Contrasting Theory to Practice: Results from Interviews with Expert Malware Analysts. Miuyin Wong, Matthew Landen, Frank Li, Fabian Monrose and Mustaque Ahamad. In USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2024.
Human Factors Malware Policy

CrashTalk: Automated Generation of Precise, Human Readable, Descriptions of Software Security Bugs. Kedrian James, Kevin Valakuzhy, Kevin Snow and Fabian Monrose. In ACM Conference and Computer and Data Security (CODASPY), June, 2024.
Malware Systems Security Measurement

Towards Practical Fabrication Stage Attacks Using Interrupt-Resilient Hardware Trojans. Athanasios Moschos, Fabian Monrose and Angelos Keromytis. In IEEE Int. Symposium on Hardware Oriented Security and Trust (HOST), May, 2024.
Hardware Systems Security Program Analysis

Improving Security Tasks Using Compiler Provenance Information Recovered At the Binary-Level. Yufei Du, Omar Alwari, Kevin Snow, Manos Antonakakis and Fabian Monrose. In ACM Conference on Computer and Communications Security (CCS), November, 2023.
Program Analysis Forensics Measurement

Stale TLS Certificates: Investigating Precarious Third-Party Access to Valid TLS Keys. Zane Ma, Aaron Faulkenberry, Thomas Papastergiou, Zakir Durumeric, Michael D. Bailey, Angelos D. Keromytis, Fabian Monrose and Manos Antonakakis. In Proceedings of ACM Internet Measurement Conference (IMC), Oct, 2023.
Measurement Networks System Security

More Carrot or Less Stick: Organically Improving Student Time Management With Practice Tasks and Gamified Assignments. Mac Malone and Fabian Monrose. In Proceedings of ACM Conference on Innovation and Technology In Computer Science Education, July, 2023.
Education Systems Security Human Factors

Securely Autograding Cybersecurity Exercises Using Web Accessible Jupyter Notebooks. Mac Malone, Yicheng Wang and Fabian Monrose. In Proceedings of ACM SIGCSE Technical Symposium, March, 2023.
Education Measurement Human Factors

Beyond the Gates: An Empirical Analysis of HTTP-managed password stealers and operators. Omar Alrawi, Athanasios Avgetidis, Kevin Valakuzhy, Charles Lever, Paul Burbage, Angelos Keromytis, Fabian Monrose, and Manos Antonakakis. In Proceedings of the USENIX Security Symposium, August 2023.
Forensics Privacy Measurement

View from Above: Exploring the Malware Ecosystem from the Upper DNS Hierarchy. Aaron Faulkenberry, Athanasios Avgetidis, Zane Ma, Omar Alrawi, Charles Lever, Panagiotis Kintis, Fabian Monrose, Angelos Keromytis, and Manos Antonakakis. In Proceedings of the Annual Computer Security Applications Conference (ACSAC), Dec 2022.
Measurement Malware Forensics

Automatic Recovery of Fine-grained Compiler Provenance Information at the Binary Level. Yufei Du, Ryan Court, Kevin Z. Snow and Fabian Monrose. USENIX Annual Technical Conference, July, 2022.

Program Analysis | Forensics | Measurement

Disentangling Style and Content for Low Resource Video Domain Adaptation: A Case Study on Keystroke Inference Attacks. John Lim, Jan-Michael Frahm and Fabian Monrose. In ACM Conference on Data and Application Security (CODASPY), April, 2022.

Biometrics | System Security | Privacy

An Online Gamified Learning Platform for Teaching Cybersecurity and More. Mac Malone, Yicheng Wang and Fabian Monrose. ACM Annual Conference on IT Education, October, 2021.

Education | Human Factors | AI/ML

DynPTA: Combining Static and Dynamic Analysis for Practical Selective Data Protection. Tapti Palit, Jarin Firose Moon, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2021.

Software Security | Systems Security | Measurement

Applicable Micropatches and Where to Find Them: Finding and Applying New Security Hot Fixes to Old Software. Mac Malone, Yicheng Wang, Kevin Snow and Fabian Monrose. IEEE International Conference on Software Testing, Verification and Validation, April, 2021.

Systems Security | Measurement | AI/ML

The Circle Of Life: A Large-Scale Study of The IoT Malware Lifecycle. Omar Alrawi, Charles Lever, Kevin Valakuzhy, Ryan Court, Kevin Snow, Fabian Monrose and Manos Antonakakis. USENIX Security Symposium, August, 2021.

Measurement | Malware | Forensics

To Gamify or Not? On Leaderboard Effects, Student Engagement, and Learning outcomes in a Cybersecurity Intervention. Mac Malone, Yicheng Wang, Kedrian James, Murray Anderegg, Jan Werner and Fabian Monrose. ACM SIGCSE Technical Symposium, 2021.

Education | Human Factors | System Security

A Flexible Framework for Expediting Bug Findings by Leveraging Past (Mis-)Behavior to Discover New Bugs. Sanjeev Das, Kedrian James, Jan Werner, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. ACM Annual Computer Security Applications Conference (ACSAC), December 2020.

Systems Security | Measurement | Forensics

Methodologies for Quantifying (Re-)randomization Security and Timing under JIT-ROP. Md Salman Ahmed, Ya Xiao, Kevin Z. Snow, Gang Tan, Fabian Monrose and Danfeng Yao. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2020.

Software Security | Systems Security | Measurement

Mitigating Data Leakage by Protecting Memory-resident Sensitive Data. Tapti Palit, Fabian Monrose and Michalis Polychronakis. In Proceedings of the Annual Computer Security Application Conference, December 2019.

Systems Security | Software Security | Privacy

The <u>SEV</u>er<u>ES</u>t Of Them All: Inference Attacks Against Secure Virtual Enclaves. Jan Werner, Joshua Mason, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. In Proceedings of the ACM Asia Conference on Computer and Communication Security, July 2019.

Systems Security | Forensics | AI/ML

SoK: The Challenges, Pitfalls, and Perils of Using Hardware Performance Counters for Security. S. Das, J. Werner, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.

Privacy Systems Security Measurement

SoK: Security Evaluation of Home-Based IoT Deployments. O. Alrawi, C. Lever, M. Antonakakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.
Privacy Systems Security Measurement

Security Concerns in Asynchronous Web Server Architectures: When Performance Optimizations Empower Data-oriented Attacks. M. Morton, J. Werner, K. Z. Snow, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE European Symposium on Security and Privacy, April, 2018.
Software Security Systems Security Measurement

Practical Attacks Against Graph-based Clustering. Y. Chen, Y. Nadji, A. Kountouras, F. Monrose, R. Perdisci, M. Antonakakis, N. Vasiloglou. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2017.
AI/ML Systems Security Measurement

Keeping Zombie Gadgets at Bay by Re-randomizing after Disclosure. Micah Morton, Forrest Li, Kevin Snow, Michalis Polychronakis and Fabian Monrose. In Proceedings of Engineering Secure Systems and Software conference (ESSoS), July, 2017.
Systems Security Measurement Software Security

Revisiting Browser Security in the Modern Era: New Data-only Attacks and Defenses. Roman Rogowski, Micah Morton, Forrest Li, Kevin Z. Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE European Symposium on Security and Privacy, March 2017.
Privacy Systems Security Software Security

Virtual U: Defeating Face Liveness Detection by Building Virtual Models From Your Public Photos. Yi Xu, True Price, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the USENIX Security Symposium, August 2016.
Systems Security Privacy AI/ML

Return to the Zombie Gadgets: Undermining Destructive Code Reads via Code-Inference Attacks. Kevin Z. Snow, Roman Rogowski, Jan Werner, Hyungjoon Koo, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2016.
Software Security Systems Security Measurement

No-Execute-After-Read: Preventing Code Disclosures in Commodity Software. Jan Werner, George Baltas, Rob Dallara, Nathan Otterness, Kevin Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the ACM Asia Conference on Computer and Communication Security, May 2016.
Systems Security Measurement Privacy

Detecting Malicious Exploit Kits using Tree-based Similarity Searches. Teryl Taylor, Xin Hu, Ting Wang, Jiyong Jang, Marc Stoeckin, Fabian Monrose and Reiner Sailer. In Proceedings of the ACM Conference on Data and Application Security and Privacy, pages 255-266, March, 2016.
Malware Privacy Measurement

Cache, Trigger, Impersonate: Enabling Context-Sensitive Honeyclient Analysis On-the-Wire. Teryl Taylor, Kevin Snow, Nathan Otterness and Fabian Monrose. In Proceedings of the 23nd ISOC Network and Distributed Systems Security Symposium (NDSS), Feb., 2016.
Systems Security Measurement Software Security

Isomeron: Code Randomization Resilient to (Just-in-Time) Return-Oriented Programming. Luca Davi, Christopher Liebchen, Ahmad-Reza Sadeghi, Kevin Z. Snow, and Fabian Monrose. In Pro-

ceedings of the Network and Distributed Systems Security Symposium (NDSS), Feb., 2015.
Systems Security Measurement Software Security

Watching the Watchers: Inferring TV Content from Outdoor Light Effusions. Yi Xu, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the 21st ACM Conference on Computer and Communications Security (CCS), November, 2014.
Systems Security Privacy AI/ML

Emergent Faithfulness to Morphological and Semantic Heads in Lexical Blends. Katherine Shaw, Elliott Moreton, Andrew White and Fabian Monrose. In Proceedings of the Annual Meeting on Phonology, February, 2014.
Phonology Human Factors Linguistics

Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, February, 2014.
Human Factors Biometrics Privacy

On the Privacy Risks of Virtual Keyboards: Automatic Reconstruction of Typed Input from Compromising Reflections. Rahul Raguram, Andrew White, Yi Xu, Jan-Michel Frahm, Pierre Georgel, and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, Volume 10, Issue 3, pages 154-167, May-June, 2013.
Privacy Systems Security Human Factors

Seeing Double: Reconstructing Obscured Typed Input from Repeated Compromising Reflections. Yi Xu, Jared Heinly, Andrew M. White, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the ACM Conference on Computer and Communications Security (CCS), November, 2013.
Privacy Systems Security Measurement

Check my profile: Leverage static analysis for fast and accurate detection of ROP gadgets. Blaine Stancill, Kevin Snow, Nathan Otterness, Fabian Monrose, Lucas Davi, and Ahmad- Reza Sadeghi. In Proceedings of the 16th International Symposium on Research in Attacks, Intrusions, and Defenses, October, 2013.
Systems Security Measurement Software Security

Crossing the Threshold: Detecting Network Malfeasance via Sequential Hypothesis Testing. Srinivas Krishnan, Teryl Taylor, Fabian Monrose and John McHugh. In Proceedings of the 42^{nd} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2013.
Systems Security Measurement AI/ML

Just-In-Time Code Reuse: On the Effectiveness of Fine-Grained Address Space Layout Randomization. Kevin Snow, Lucas Davi, Alexandra Dmitrienko, Christopher Liebchen, Fabian Monrose and Ahmad-Reza Sadeghi. In Proceedings of 34^{th} IEEE Symposium on Security and Privacy, May, 2013. (**Best Student Paper Award**).
Privacy Systems Security AI/ML

Clear and Present Data: Opaque Traffic and its Security Implications for the Future. Andrew White, Srinivas Krishnan, Michael Bailey, Fabian Monrose and Phil Porras. In Proceedings of the 20^{th} ISOC Network and Distributed Systems Security Symposium, Feb., 2013.
Systems Security Measurement AI/ML

Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm, Fabian Monrose and Paul van Oorschot. In Proceedings of the 21^{th} USENIX Security Symposium, August, 2012.

Human Factors Biometrics Privacy

Toward Efficient Querying of Compressed Network Payloads. Teryl Taylor, Scott E. Coull, Fabian Monrose and John McHugh. In USENIX Annual Technical Conference, June, 2012.
Measurement Systems Security AI/ML

Trail of Bytes: New Techniques for Supporting Data Provenance and Limiting Privacy Breaches. Srinivas Krishnan, Kevin Snow, and Fabian Monrose. IEEE Transactions on Information Forensics and Security, pages 1876-1889, Issue 6, Volume 7, 2012.
Forensics Privacy Human Factors

Understanding Domain Registration Abuses (Invited Paper). Scott Coull, Andrew White, Ting-Fang Yen, Fabian Monrose and Michael K. Reiter. Special Issue of Computers & Security (COSE), pages 806-815, Volume 31, Number 7, October, 2012.
Measurement Systems Security AI/ML

iSpy: Automatic Reconstruction of Typed Input from Compromising Reflections. Rahul Raguram, Andrew White, Dibyendusekhar Goswami, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the 18^{th} ACM Conference on Computer and Communications Security (CCS), November, 2011.
Privacy Systems Security Measurement

ShellOS: Enabling Fast Detection and Forensic Analysis of Code Injection Attacks. Kevin Snow, Srinivas Krishnan, Fabian Monrose and Niels Provos. In Proceedings of the 20^{th} USENIX Security Symposium, August, 2011.
Forensics Systems Security Software Systems

An Empirical Study of the Performance, Security and Privacy Implications of Domain Name Prefetching. Srinivas Krishnan and Fabian Monrose. In Proceedings of the 41^{st} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Dependable Computing and Communications Symposium, June, 2011.
Measurement Systems Security Software Security

Amplifying Limited Expert Input to Sanitize Large Network Traces. Xin Huang, Fabian Monrose, and Michael K. Reiter. In Proceedings of the 41^{st} Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2011.
Systems Security Measurement AI/ML

Phonotactic Reconstruction of Encrypted VoIP Conversations. Andrew White, Kevin Snow, Austin Matthews and Fabian Monrose. In Proceedings of 32^{nd} IEEE Symposium on Security and Privacy, May, 2011. (**Best Paper Award**).
Privacy System Security Measurement

Towards Optimized Probe Scheduling for Active Measurement Studies. Daniel Kumar, Fabian Monrose and Michael K. Reiter. In Proceedings of the 6^{th} International Conference on Internet Monitoring and Protection (ICIMP), March, 2011 (**Best Paper Award**).
Measurement Privacy Networks

On Measuring the Similarity of Network Hosts: Pitfalls, New Metrics, and Empirical Analyses. Scott Coull, Micheal Bailey and Fabian Monrose. In Proceedings of the 18^{th} Annual Network and Distributed Systems Security Symposium, February, 2011.
Systems Security Measurement AI/ML

Trail of Bytes: Efficient Support for Forensic Analysis. Srinivas Krishnan, Kevin Snow, and Fabian Monrose. In Proceedings of the 17^{th} ACM Conference on Computer and Communications

Security (CCS), November, 2010.
Forensics Systems Security Software Systems

The Security of Modern Password Expiration: An Algorithmic Framework and Empirical Analysis. Yinqian Zhang, Fabian Monrose, and Michael K. Reiter. In Proceedings of the 17^{th} ACM Conference on Computer and Communications Security (CCS), November, 2010.
Human Factors Measurement AI/ML

Traffic Classification using Visual Motifs: An Empirical Evaluation. Wilson Lian, Fabian Monrose and John McHugh. In Proceedings of the 7^{th} International Symposium on Visualization for Cyber Security (VizSec), September, 2010.
Measurement AI/ML Systems Security

Understanding Domain Registration Abuses. Scott Coull, Andrew White, Ting-Fang Yen, Fabian Monrose and Michael K. Reiter. In Proceedings of the International Information Security Conference, September, 2010.
Measurement Systems Security AI/ML

English Shellcode. Josh Mason, Sam Small, Greg McManus and Fabian Monrose. In Proceedings of the 16^{th} ACM Conference on Computer and Communications Security (CCS), pages 524-533, November, 2009.
Systems Security Software Security AI/ML

Browser Fingerprinting from Coarse Traffic Summaries: Techniques and Implications. Ting-Fang Yen, Xin Huang, Fabian Monrose and Michael K. Reiter. In Proceedings of 6^{th} Conference on Detection of Intrusions and Malware and Vulnerability Assessment, pages 157-175, July, 2009.
Malware Measurement AI/ML

Toward Resisting Forgery Attacks via Pseudo-Signatures. Jin Chen, Dan Lopresti and Fabian Monrose. In Proceedings of 10^{th} International Conference on Document Anaalysis and Recognition (ICDAR), July, 2009.
Systems Security Privacy Human Factors

The Challenges of Effectively Anonymizing Network Data. Scott Coull, Fabian Monrose, Michael K. Reiter and Michael Bailey. In Proceedings of the DHS Cybersecurity Applications and Technology Conference for Homeland Security (CATCH), pages 230-236, 2009.
Measurement Systems Security Networks

Efficient Defenses Against Statistical Traffic Analysis. Charles Wright, Scott Coull and Fabian Monrose. In Proceedings of Network and Distributed Systems Security, pages 237-250 Feb, 2009.
Systems Security Measurement AI/ML

Towards Practical Biometric Key Generation with Randomized Biometric Templates. Lucas Ballard, Seny Kamara, Fabian Monrose, and Michael K. Reiter. In Proceedings of the 15^{th} ACM Conference on Computer and Communications Security, pages 235-244. Oct, 2008.
Biometrics Crypto Measurement

All Your iFrames point to us: Characterizing the new malware frontier. Niels Provos, Panayiotis Mavrommatis, Moheeb Rajab, and Fabian Monrose. In Proceedings of the 17^{th} USENIX Security Symposium, pages 1-15, July, 2008.
Malware Systems Security Forensics

To Catch a Predator: A Natural Language Approach for Eliciting Protocol Interaction. Sam Small, Josh Mason, Fabian Monrose, Niels Provos and Adam Stubblefield. In Proceedings of the 17^{th} USENIX Security Symposium, pages 171-183, July, 2008.

Privacy Systems Security Networks

Peeking Through the Cloud: DNS-based client estimation techniques and its applications. Moheeb Rajab, Niels Provos, Fabian Monrose and Andreas Terzis. In Proceedings of the 6^{th} Applied Cryptography and Network Security conference (ACNS), pages 21-38, June, 2008.
Privacy Measurement Software Systems

Spot Me If You Can: recovering spoken phrases in encrypted VOIP conversations. Charles Wright, Lucas Ballard, Scout Coull and Fabian Monrose. In Proceedings of 29^{th} IEEE Symposium on Security and Privacy, May, 2008 (17 pages).
Biometrics Privacy Measurement

Taming the Devil: Techniques for Evaluating Anonymized Network Data. Scott Coull, Charles Wright, Angelos Keromytis, Fabian Monrose, and Michael Reiter. In Proceedings of the 15^{th} Annual Network and Distributed Systems Security Symposium, pages 125-146, Feb., 2008.
Systems Security Measurement Networks

On Web Browsing Privacy in Anonymized NetFlows. Scott Coull, Michael Collins, Charles Wright, Fabian Monrose, and Michael Reiter. In Proceedings of the 16^{th} USENIX Security Symposium, pages 339-352, August, 2007.
Privacy Systems Security Human Factors

Language Identification of Encrypted VoIP Traffic: Alejandra y Roberto or Alice and Bob? Charles Wright, Lucas Ballard, Fabian Monrose and Gerald Masson. In Proceedings of the 16^{th} USENIX Security Symposium, pages 43-54, August, 2007.
Crypto Measurement AI/ML

Towards Valley-free Inter-domain Routing. Sophie Qiu, Patrick McDaniel and Fabian Monrose. In Proceedings of the IEEE International Conference on Communications, June, 2007. (8 pages)
Measurement Systems Security AI/ML

Playing Devil's Advocate: Inferring Sensitive Information from Anonymized Traces. Scott Coull, Charles Wright, Fabian Monrose, Michael Collins and Michael Reiter. In Proceedings of the 14^{th} Annual Network and Distributed Systems Security Symposium (NDSS), pages 35-47, February 2007.
Systems Security Measurement Networks

A Multifaceted Approach to Understanding the Botnet Phenomenon. Jay Zarfoss, Moheeb Rajab, Fabian Monrose, and Andreas Terzis. In Proceedings of the ACM SIGCOMM/USENIX Internet Measurement Conference (IMC), pages 41-52, October, 2006.
Measurement Malware AI/ML

Fast and Evasive Attacks: Highlighting the Challenges Ahead. Moheeb Rajab, Fabian Monrose, and Andreas Terzis. In Proceedings of the 9^{th} International Symposium on Recent Advances in Intrusion Detection (RAID), pages 206-225, September, 2006.
Malware Systems Security Measurement

Biometric Authentication Revisited: Understanding the Impact of Wolves in Sheep's Clothing. Lucas Ballard, Fabian Monrose, and Daniel Lopresti. In Proceedings of the USENIX Security Symposium, pages 29-41, August 2006.
Biometrics Human Factors Privacy

On Origin Stability in Inter-Domain Routing. Sophie Qui, Patrick McDaniel, Fabian Monrose and Aviel D. Rubin. In Proceedings of IEEE International Symposium on Computers and Communications (ISCC), pages 489-496, July, 2006.
Measurement Systems Security Networks

Memory Bound Puzzles: A Heuristic Approach. Sujata Doshi, Fabian Monrose and Aviel D. Rubin. In Proceedings of the International Conference on Applied Cryptography and Network Security (ACNS), pages 98-113, June 2006.
Crypto | Systems Security | Measurement

Achieving Efficient Conjunctive Searches on Encrypted Data. Lucas Ballard, Seny Kamara and Fabian Monrose. In Proceedings of 7^{th} International Conference on Information and Communications Security (ICICS), pages 414-426, December, 2005.
Crypto | Systems Security | Measurement

On the Effectiveness of Distributed Worm Monitoring. Moheeb Rajab, Fabian Monrose and Andreas Terzis. In Proceedings of 14^{th} USENIX Security Symposium, pages 225-237, August, 2005.
Measurement | Systems Security | Networks

Scalable VPNs for the Global Information Grid. Bharat Doshi, Antonio De Simone, Fabian Monrose, Samuel Small, and Andreas Terzis. In Proceedings of IEEE MILCOM, May, 2005. (7 pages)
Systems Security | Measurement | Networks

An Extensible Platform for Evaluating Security Protocols. Seny Kamara, Darren Davis, Ryan Caudy and Fabian Monrose. In Proceedings of the 38^{th} Annual IEEE Simulation Symposium (ANSS), pages 204-213, April, 2005.
Systems Security | Networks | Software Security

Efficient Time Scoped Searches on Encrypted Audit Logs. Darren Davis, Fabian Monrose, and Michael Reiter. In Proceedings of the 5^{th} International Conference on Information and Communications Security (ICICS), pages 532-545, October, 2004.
Crypto | Systems Security | Measurement

On User-Choice in Graphical Password Systems. Darren Davis, Fabian Monrose, and Michael Reiter. In Proceedings of the 13^{th} USENIX Security Symposium, pages 151-164, August, 2004.
Human Factors | Systems Security | Privacy

Toward Speech-Generated Cryptographic Keys on Resource Constrained Devices. Fabian Monrose, Micheal Reiter, Daniel Lopresti, Chilin Shih, and Peter Li. In Proceedings of the 11^{th} USENIX Security Symposium, pages 283–296, August, 2002.
System Security | Biometrics | Privacy

Using Voice to Generate Cryptographic Keys: A Position Paper. Fabian Monrose, Michael Reiter, Peter Li and Susanne Wetzel. In Proceedings of the 2001: A Speaker Odyssey workshop, pages 237-242, 2001.
Privacy | Biometrics | System Security

Cryptographic Key Generation from Voice. Fabian Monrose, Michael Reiter, Peter Li and Susanne Wetzel. In Proceedings of the 21^{st} Annual IEEE Symposium on Security & Privacy, pages 202-212, May 2001.
Privacy | Biometrics | System Security

Privacy-Preserving Global Customization. Bob Arlein, Ben Jai, Markus Jakobsson, Fabian Monrose, and Michael Reiter. In Proceedings of the 2^{nd} ACM Conference on Electronic Commerce, pages 176-184. April 2000.
Privacy | Systems Security | Human Factors

Password Hardening using Keystroke Dynamics. Fabian Monrose, Michael K. Reiter and Susanne Wetzel. In Proceedings of the 6^{th} ACM Conference on Computer and Communications Security, pages 73-82, November 1999.

System Security Human Factors Measurement

The Design and Analysis of Graphical Passwords. Ian Jermyn, Alain Mayer, Fabian Monrose, Michael K. Reiter and Aviel D. Rubin. In Proceedings for the 8^{th} USENIX Security Symposium, pages 1-14, August 1999. (**Best Paper Award**).
Privacy Systems Security Human Factors

Distributed Execution with Remote Audit. Fabian Monrose, Peter Wyckoff and Aviel D. Rubin. In Proceedings of the ISOC Network and Distributed Systems Security Symposium (NDSS), pages 103-113, February 1999.
Systems Security Measurement Software Security

Third Party Validation for Java Applications. Ian Jermyn, Fabian Monrose, and Peter Wyckoff. In Proceedings of the International Conference on Computers and their Applications, 1998.
Systems Security Measurement Software Security

Authentication via Keystroke Dynamics. Fabian Monrose and Aviel D. Rubin. In Proceedings of the 4^{th} ACM Conference on Computer and Communications Security, pages 48-56, April 1997.
Biometrics Human Factors System Security

Teaching

- *Advanced Computer Security*, Fall 2024, 8 students.
- *Advanced Computer Security*, Fall 2024, 14 students.
- *Advanced Computer Security*, Fall 2023, 16 students.
- *Introduction to Computer Security*, Spring 2022, 59 students.
- *Computer and Network Forensics*, Spring 2021, 13 students.
- *Technical Communications and Writing in Computer Science*, Spring 2021, 14 students.
- *Introduction to Computer Security*, Fall 2020, 19 students.
- *Technical Communications and Writing in Computer Science*, Spring 2020, 18 students.
- *Introduction to Computer Security*, Spring 2019, 34 students.
- *Introduction to Computer Security*, Spring 2017, 48 students.
- *Computer Forensics*, Fall 2016, 14 students.
- *Introduction to Computer Security*, Spring 2016, 43 students.
- *Introduction to Computer Security*, Spring 2015, 30 students.

Students

- Sophie Qui (co-advised with G. Mason), Ph.D., Spring 2007, (*Cisco Systems*)
- Charles Wright, Spring 2008, Ph.D., (*Portland State University*)
- Seny Kamara, Spring 2008, Ph.D., (*Brown University*)
- Moheeb Abu Rajab (co-advised with A. Terzis), Ph.D., Spring 2008, (*Google*)
- Lucas Ballard, Spring 2008, Ph.D., (*Google*)
- Scott Coull, Fall 2009, Ph.D., (*RedJack*)
- Josh Mason, Fall 2009, Ph.D., (*Research Professor, UIUC*)
- Kevin Snow, Spring 2014, Ph.D., *Zeropoint Dynamics*

- Andrew White, Fall 2015, Ph.D., *Netflix*
- Teryl Taylor, Summer 2016, Ph.D., *IBM TJ Watson*
- Srinivas Krishnan (co-advised with K. Jeffay), *Google*
- Yi Xu, (co-advised with Jan-Michael Frahm), Spring 2016, Ph.D. *Snapchat*
- John Lim (co-advised with Jan-Michael Frahm) *Salesforce*
- Mac Malone, Fall 2023, Ph.D. (*Lean*)
- Kedrian James, Spring 2024, Ph.D. (*Highpoint University*)
- Yufei Du, expected graduation: Fall 2026
- Kevin Valakuzhy, expected graduation: Fall 2026
- Tillson Galloway, expected graduation: Fall 2026
- Vinny Adjibi, expected graduation: Spring 2028
- James Fullwood, expected graduation: Summer 2028
- Lydia Han, expected graduation: Fall 2028

Recent Doctoral Committees

(**Advisor**) Kevin Z. Snow, *Identifying Code Injection and Code Reuse Payloads in Memory Error Exploits*, Ph.D., University of North Carolina at Chapel Hill, Fall, 2014.

(**Advisor**) Andrew M. White, *Practical Analysis of Encrypted Network Traffic*, Ph.D., University of North Carolina at Chapel Hill, Fall, 2015.

(**Co-Advisor**) Yi Xu, *Toward Robust Video Event Detection and Retrieval Under Adversarial Constraints*, Ph.D., University of North Carolina at Chapel Hill, Spring, 2016.

(**Advisor**) Teryl Taylor, *Using Context to Improve Network-based Malware Detection*, Ph.D., University of North Carolina at Chapel Hill, Summer, 2016.

(**Reader**) Dennis Bueno, *Automated Software Security Analysis Using Control and Data Abstractions*, Ph.D., University of Michigan, 2017.

(**Reader**) Tracey John, *Virtual Career Advisor System*, MPhil., University of the West Indies, Barbados, 2017.

(**Reader**) Louis Soleyn, *Software Techniques for Implementing Dynamic, Network-Aware, Energy-Efficient Mobile Applications*, MPhil., University of the West Indies, Barbados, 2018.

(**Reader**) Chaz Lever, *Empirical Analysis of Existing and Emerging Threats at Scale Using DNS*, Georgia Institute of Technology, Fall, 2017.

(**Reader, external examiner**) Adrian Tang, *Security Engineering of Hardware-Software Interfaces*, Columbia University, Spring, 2018.

(**Reader**) Calvin Deutschbein, *Mining Secure Behavior of Hardware Designs*, UNC Chapel Hill, July, 2021.

(**Reader**) Tapti Palit, *Sensitive Data Encryption: A Scalable Defense against Sensitive Data Leakage*, Stony Brook University, October, 2021.

(**Reader, external examiner**) Salman Ahmed, *Quantitative Metrics and Measurement Methodologies for System Security Assurance*, Virginia Tech, Dec, 2021.

(**Reader**) Omar Alrawi, *Security and Threat Evaluation of Smart Home Technology*, Georgia Institute of Technology, July, 2022.

(**Advisor**) Mac Malone, *Improving Student Engagement and Learning Outcomes with a Gamified Learning Platform*, Ph.D., University of North Carolina at Chapel Hill, May, 2023.

(**Reader**) Evan Downing, *Improving the Understanding of Malware Using Machine Learning*, Georgia Institute of Technology, December 2023.

(**Advisor**) Kedrian James, *Improving Software Security Triage and Remediation*, Ph.D., University of North Carolina at Chapel Hill, May, 2024.

(**Reader**) Kevin Hutto, *Remote Sensor Security Through Encoded Computation and Cryptographic Signatures*, Ph.D., Georgia Institute of Technology, May, 2024.

(**Reader**) Miuyin M. Wong, *Narrowing the Gap between Research and Practice through the Understanding of Malware Analysis Workflows*, Georgia Institute of Technology, 2024

(**Reader**)Athanasios Avgetidis, *Towards Understanding the Lifecycle of Malicious Network Infrastructure*, Georgia Institute of Technology, 2025

(**Reader**) Aaron Faulkenberry, *Internet-Scale Discovery and Measurement of DNS Monitoring for Cyber Threats*, Georgia Institute of Technology, 2025

BS/MS
Advisees

Wilson Lian, *Traffic Classification using Visual Motifs*, University of North Carolina at Chapel Hill, Honors Thesis, 2010.

Austin Matthews, *Phonetic Edit Distance: New Techniques and Empirical Analyses*, University of North Carolina at Chapel Hill, Honors Thesis, 2011.

Chris Allen, *WarFlying: Creating Aerial WiFi-maps using Custom Drones*, University of North Carolina at Chapel Hill, Honors Thesis, 2013.

Rob Dallara, *On the (In)effectiveness of Execute-no-Read (XNR) as a Defense against Just-in-Time Code Reuse Attacks.* University of North Carolina at Chapel Hill, Comprehensive paper, 2015.

Micah Moreton, *Detecting Potential for Server-side Data-oriented Attacks*, University of North Carolina at Chapel Hill, 2017

Roman Rogowski, *Revisiting Browser Security in the Modern Era*. University of North Carolina at Chapel Hill, Comprehensive paper, 2017.

Boo Fullwood, *Revisiting Charger-based Side Channel Password Inference Attacks*. University of North Carolina at Chapel Hill, Comprehensive paper, 2022.

Kevin Lane, *Cloud-based Architectures for Detecting VoIP Scams*. University of North Carolina at Chapel Hill, Comprehensive paper, 2022.

Recent Service
(*last 5 years***)**

- ECE PhD Admissions committee (Area Chair for Software & Systems), 2023-2026
- 32^{nd} ACM Conference on Computer and Communications Security, 2026
- 47^{th}IEEE Symposium on Security and Privacy, 2026
- 32^{nd} ACM Conference on Computer and Communications Security, 2025

- 14^{th} ACM Conference on Data and Application Security and Privacy, 2024
- Faculty Recruiting Committee member, School of Cybersecurity & Privacy, 2023
- 13^{th} ACM Conference on Data and Application Security and Privacy, 2023

Recent Patents/Filings

Kevin Valakuzhy, Miuyin Yong Wong, Fabian Monrose, Douglas Blough and Mustaque Ahmad. LLM-Enabled System for Locating and Bypassing Evasive Malware Tactics, Provision Patent Filing 165972-00196, October 14, 2025.

Kevin Valakuzhy, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. Systems and Methods for Malware Similarity via Datastore of Assembly-level Malicious Behaviors Extracted from Memory, US Patent Application No. 19/380,650, November, 2025.

Jan Werner, Rob Dallara, George Baltas, Michalis Polychronakis, Kevin Snow and Fabian Monrose. *Methods, Systems, and Computer Readable Media for Preventing Code Reuse Attacks.* U.S. Patent No. 10,628,589, April, 2020.

Xin Hu, Jiyong Jang, Fabian Monrose, Marc Philippe Stoecklin, Teryl Taylor and Ting Wang. *Detecting Web Exploit Kits by Tree-based Structural Similarity Search*. U.S. Patent 10,560,471, Feb, 2020.

T. Taylor, K. Snow, N. Otterness, and F. Monrose. *Methods, systems, and computer readable media for detecting malicious network traffic*, U.S. Patent No. 9,992,217, June 2018.

Srinivas Krishnan, Fabian Monrose, Michael Bailey, Phillip Porras, and Andrew White. *Methods, Systems, and Computer Readable Media for Rapid Filtering of Opaque Data Traffic*, U.S. Patent No. 9,973,473, May 2018.

Srinivas Krishnan, Teryl Taylor, Fabian Monrose and John McHugh. *Methods, Systems, and Computer Readable Media for Detecting Compromised Computing Hosts*. U.S. Patent No. 9,934,379, April 2018.

Srinivas Krishnan, Kevin Snow, and Fabian Monrose. *Methods, Systems, and Computer Readable Media for Efficient Computer Forensic Analysis and Data Access Control*. U.S. Patent No. 9,721,089, August 2017.

Kevin Z. Snow, Fabian Monrose and Srinivas Krishnan. *Methods, Systems, and Computer Readable Media for Detecting Injected Machine Code*. U.S. Patent 9,305,165, April 5, 2016.

Fabian Monrose, Teryl Taylor, Srinivas Krishnan and John McHugh. United States Patent Application Serial No. 14/773,660 for *Methods, Systems, and Computer Readable Media for Detecting a Compromised Computing Host*, March, 2015.